OMB No. 3145-0279 (Expiration Date 10/31/2026)

NIH BIOGRAPHICAL SKETCH COMMON FORM

Name: Cherian, Aaron T

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0003-4944-2814

Position Title: Chief Scientific Officer

Organization and Location: FreeMoCap Foundation, Boston, Massachusetts, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
Northeastern University, Boston, Massachusetts, United States	Doctor of Philosophy (PHD)	09/2021	05/2026	Bioengineering
Brown University, Providence, Rhode Island, United States	Master of Science (MS)	09/2018	05/2020	Biomedical Engineering
University of Maryland College Park, College Park, Maryland, United States	Bachelor of Science (BS)	09/2014	05/2018	Bioengineering

Appointments and Positions

2026 - present Chief Scientific Officer, FreeMoCap Foundation, Boston, Massachusetts, United States

2021 - 2026 Graduate Research Assistant, Northeastern University, Bioengineering Department, Boston, Massachusetts, United States

Products

Products Closely Related to the Proposed Project

1. Cherian AT. Open-Source Development and Validation of a Low-Cost Markerless System for Quantitative Motion Analysis. Boston, Massachusetts: Northeastern University; 2026 May. Available from: https://hdl.handle.net/2047/D20859645
2. Jonathan Samir Matthis, Philip Queen, Aaron Cherian, Trent Wirth, Endurance Idehen. FreeMoCap: A free, open source markerless motion capture system. 2026 August. Available from: https://zenodo.org/doi/10.5281/zenodo.22131400 DOI: 10.5281/ZENODO.22131400
3. Aaron Cherian. FreeMoCap Validation Dataset: Multi-Camera Markerless Motion Capture of Treadmill Gait and Standing Balance. Zenodo; 2026 September. Available from: https://zenodo.org/doi/10.5281/zenodo.23041223 DOI: 10.5281/ZENODO.23041223
4. Aaron Cherian, jonmatthis. SkellyForge: The 3D data processing backend for the FreeMoCap Project. 2023 December. Available from: https://zenodo.org/doi/10.5281/zenodo.10395649 DOI: 10.5281/ZENODO.10395649
5. jonmatthis, Endurance Idehen, philipqueen, Mac Prible, Aaron Cherian. SkellyCam: The camera backend for the FreeMoCap software. 2026 April. Available from: https://zenodo.org/doi/10.5281/zenodo.19921107 DOI: 10.5281/ZENODO.19921107

Other Significant Products Highlighting Contributions to Science

1. Nie J, Eom K, AlGhosain H, Neifert A, Cherian A, Gerbaka G, Ma K, Liu T, Lee J. Intravitreally Injected Plasmonic Nanorods Activate Bipolar Cells with Patterned Near-Infrared Laser Projection. ACS Nano. 2025 March 20; 19(12):11823-11840. Available from: https://pubs.acs.org/doi/10.1021/acsnano.4c14061 DOI: 10.1021/acsnano.4c14061; NIHMSID: NIHMS2081004
2. philipqueen, jonmatthis. SkellyTracker: The image analysis backend for the FreeMoCap Project. 2023 December. Available from: https://zenodo.org/doi/10.5281/zenodo.10395641 DOI: 10.5281/ZENODO.10395641
3. Aaron Thomas Cherian. Development and Optimization of an Intrinsic Optical Imaging System for Hemodynamic Correction of Fluorescent Signals. 2020 May. Available from: https://repository.library.brown.edu/studio/item/bdr:1129219/ DOI: 10.26300/XHJ5-KV66
4. philipqueen, jonmatthis, Aaron Cherian. SkellyClicker: For labeling and traing multi-camera video datasets. 2026 April. Available from: https://zenodo.org/doi/10.5281/zenodo.19389889 DOI: 10.5281/ZENODO.19389889

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§287, 1001, 1031 and 31 U.S.C. §§3729-3733 and 3802.

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NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Cherian, Aaron T

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0003-4944-2814

Position Title: Chief Scientific Officer

Organization and Location: FreeMoCap Foundation, Boston, Massachusetts, United States

Personal Statement

I am a researcher and Chief Scientific Officer of the FreeMoCap Foundation, where my work focuses on the development, validation, and application of accessible, open-source motion capture technologies. I received my PhD in Bioengineering from Northeastern University in 2026, where my doctoral research focused on developing and evaluating a low-cost, multi-camera markerless motion capture system for quantitative movement analysis. This work involved comparing markerless reconstruction methods against marker-based reference measurements and evaluating their performance for biomechanical outcomes, including gait kinematics, spatiotemporal parameters, and standing balance measures.

My research background combines biomedical instrumentation, quantitative data analysis, and scientific software development. During my master's research at Brown University, I developed and optimized an optical imaging system for neuroscience research, gaining experience in experimental measurement and biological signal analysis. My earlier research at Burke Neurological Institute also introduced me to applications of quantitative measurement in rehabilitation. More recently, I have contributed to the development of FreeMoCap and its supporting software, including tools for three-dimensional motion data processing and multi-camera recording.

I recently released an open validation dataset containing markerless and marker-based motion capture measurements collected during walking and standing balance tasks, together with analysis resources supporting reproducibility. This work has highlighted the importance of documenting measurement definitions, processing methods, and data provenance when interpreting and comparing biomechanical outcomes.

I am well suited to lead the proposed DAPR pilot project because it builds directly on these experiences. The project will harmonize existing motion capture datasets and extend FreeMoCap to support structured metadata capture, storage, and export. My experience with multiple pose estimation methods, biomechanical outcome measures, scientific software, and publicly shared research data provides a foundation for developing and evaluating these capabilities in collaboration with the DAPR Resource Core.

Honors

2016	Design Cultures and Creativity Capstone Award, University of Maryland
2016	Honors College Citation, University of Maryland
2014	National Merit Siemens Scholarship, National Merit Scholarship Corporation
2014 - 2018	Presidential Scholarship, University of Maryland

Contributions to Science

1. Development and maintenance of accessible, open-source motion capture

   Cost and technical complexity can limit access to quantitative movement analysis. I contribute to the development and maintenance of FreeMoCap, an open-source markerless motion capture system that uses synchronized consumer cameras to reconstruct three-dimensional human movement. My work spans the main FreeMoCap application and its supporting software libraries, including tools for camera recording, pose estimation, three-dimensional reconstruction, and biomechanical data processing.

   My contributions include developing features and analysis tools, resolving software defects and user-reported issues, managing releases and dependencies, and improving testing and application reliability. Together, these activities support FreeMoCap as a usable and adaptable research platform. Its open-source implementation allows researchers to inspect the methods used to generate movement measurements, modify processing workflows, and build on existing tools. My

development and maintenance work helps sustain these capabilities as the software evolves, supporting broader access to quantitative movement research and reproducible analysis.

2. Validation and reproducibility of markerless biomechanical measurements

   The usefulness of markerless motion capture depends on the accuracy of the measurements derived from it, which can vary across tracking methods, tasks, and outcomes. In my doctoral research, I evaluated FreeMoCap reconstructions using MediaPipe, RTMPose, and ViTPose against simultaneously recorded Qualisys marker-based measurements during treadmill walking and standing balance. The comparisons showed that performance depended on both the pose estimator and the measurement being evaluated. Methods that performed well for gait did not necessarily perform best for standing balance, demonstrating the importance of validating specific outcomes for their intended use.

   I developed the analysis workflows and evaluated agreement in joint kinematics, spatiotemporal gait measures, and center-of-mass-based balance measures. I also released the associated validation dataset and reproducible analysis resources, making matched markerless and marker-based trajectories available for independent reanalysis and method development. Together, these contributions provide evidence for interpreting FreeMoCap measurements and a reusable basis for evaluating changes to motion-capture and analysis methods.

3. Adapting markerless motion capture to prosthetic gait assessment

   General-purpose pose-estimation models can misidentify landmarks on prosthetic limbs, limiting their use for quantitative gait assessment. In a proof-of-concept study involving one participant with a transfemoral amputation, I developed and evaluated a hybrid approach combining RTMPose whole-body estimates with a DeepLabCut model trained to identify prosthetic landmarks. Comparisons with Qualisys marker-based measurements showed that the hybrid approach reduced prosthetic landmark error and recovered changes in selected gait measures across controlled prosthetic alignment adjustments. Performance remained dependent on the outcome, identifying limitations as well as improvements.

   My contributions included model development, integration of prosthetic landmarks into the reconstruction workflow, and biomechanical comparisons with the reference measurements. This work demonstrates how an open-source motion-capture workflow can be adapted to anatomical and device-related features that general-purpose models may not represent adequately. It provides a basis for further evaluation of markerless methods for prosthetic gait assessment. A manuscript describing this work is under revision.

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